

Chemistry

A. Short Question

1. What is Chemical bond?

A chemical is the association of atoms or ions to form molecules, crystals or other structures.

2. What is ionic/covalent/coordination bond?

Ionic bond: The chemical bond which is formed by the transference of electron from one atom to other is called ionic or electrovalent bond

Covalent bond: The covalent bond is formed by the sharing of electrons by both the atoms equally. This formed compound is called covalent compound.

Coordination bond: The covalent bond which forms by the sharing of both electrons (electron pair) from one atom to another atom is called coordination bond or coordination covalent bond.

3. What is hybridization?

Hybridization is the process in which two or more atomic orbitals of valence shell of an atom mix up to form the same number of identical orbitals of equal energy.

OR

The phenomenon of mixing up of atomic orbitals of similar energies and formation of equivalent number of entirely new orbitals of identical shape and energy is known as "hybridization" (and the new orbitals so formed is called as "hybrid orbitals")

4. What is sp/sp²/sp³ hybridization?

sp-hybridization: The combination of one s and one p-orbital to form two hybrid orbitals of equal energy is known as sp-hybridization.

sp²-hybridization: The combination of one s and two p-orbitals to form three hybrid orbitals of equal energy is known as sp²- hybridization. (usually 120° Trigonal Planer)

sp³-hybridization: The combination of one s and three p-orbitals to form four hybrid orbitals of equal energy is known as sp²- hybridization. (usually 109.5° Tetrahedral)

5. What is VSEPR theory?

The Valence-Shell-Electrons-Pair-Repulsion Theory (VSEPR), proposes that the stereochemistry of an atom in a molecule is determined primarily by the repulsive interactions among all the electron pairs in its valence shell.

6. What transition elements?

d-block elements which form at least one ion with partially filled sub-shell of d electrons i.e., d¹ – d⁹, are called transition elements or transition metals.

7. Why all d-block elements aren't transition elements?

Some d-block elements have completely filled d orbitals like Zn, Hg etc. which do not enable them to have variable oxidation states, but transition elements always have partially filled d orbitals which enable them to have variable oxidation states. Hence all d-block elements are not transition elements.